Introduction To Industry 4.0 And Industrial Internet Of Things

Assignment-10

Number of questions: 15 **Total marks: 15 X 1= 15**

1. Which of the following is not a part of the SDIIoT Architecture?

    a. WSN
    b. Private network
    c. Industrial cloud
    d. Industrial bus and network

Answer: (b)

Justification: Please refer to Slide 2 of Lecture 1 of Week 10

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2. Which of the following is not a part of the SDWSN platform in the context of Industry 4.0?

    a. Applications
    b. Data plane
    c. Control plane
    d. Terrestrial plane

Answer: (d)

Justification: Please refer to Slide 3 of Lecture 1 of Week 10

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3. Data processing is generally performed in _____ present in the SDIIoT architecture?

    a. LAN
    b. Public network
    c. Industrial cloud
    d. None of the above

Answer: (c)

Justification: Please refer to Slide 5 of Lecture 1 of Week 10

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4. R stands for _____ in STRIDE Threat Model of IoT security. Fill in the blanks.

a. Repudiation
b. Reliable
c. Replicable
d. Resonance

Answer: (a)

Justification: Please refer to Slide 10 of Lecture 2 of Week 10

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5. Application layer of the IoT attack surface is vulnerable to which of the following IIoT Attack Vectors?

a. Reversal attack
b. Man-in-the-middle attack
c. Jamming attack
d. None of the above

Answer: (a)

Justification: Please refer to Slide 12 of Lecture 2 of Week 10

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6. What does trust permeation aim to establish within the IIoT system?

a. Unit hierarchies
b. Device integrity
c. Manufacturing logistics
d. Development life cycle

Answer: (b)

Justification: Please refer to Slide 16 of Lecture 2 of Week 10

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7. Which of the following does not represent the building blocks of IIoT security framework?

a. Fog device protection
b. Cloud security
c. Data protection
d. Embezzlement protection

Answer: (d)

Justification: Please refer to Slide 4 of Lecture 3 of Week 10

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8. Which of the following standards is supported by the OT security of IIoT?

a. ISO/IEC 154083
b. ISO 27017, NIST SP 800-144, ENISA standard
c. NERC-CIP
d. ISO series of standards for privacy, framework and regulations

Answer: (c)

Justification: Please refer to Slide 21 of Lecture 3 of Week 10

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9. Which of the following is not a key component of cloud security of IIoT framework?

a. Data privacy
b. Flow-rule protection
c. Access control
d. Identity management

Answer: (b)

Justification: Please refer to Slide 10 of Lecture 3 of Week 10

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10. Which of the following is not a key feature of smart factory?

a. Connected
b. Optimized
c. Ambiguous
d. Agile

Answer: (c)

Justification: Please refer to Slide 5 of Lecture 4 of Week 10

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11. Which of the following is incorrect about Kuka – an IoT-Enabled Factory?

a. A German robotics maker
b. Built an IoT-enabled factory
c. Robots are connected with a private cloud

d. Deployed on the A330 and A350 final assembly lines in Toulouse

Answer: (d)

Justification: Please refer to Slide 7 of Lecture 4 of Week 10

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12. _____ is carried out by Amazon – Robotic Shelves. Fill in the blanks.

a. Automated product search
b. Manufacturing of drones
c. AR app based end-to-end factory floor view
d. None of the above

Answer: (a)

Justification: Please refer to Slide 11 of Lecture 4 of Week 10

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13. Which of the following is incorrect about Maersk – Intelligent Shipping?

a. It is a container shipping company
b. Uses IoT for preserving refrigerated containers
c. Does not use blockchain technology for supply chain optimization
d. Tracks assets and fuel consumption using sensors

Answer: (c)

Justification: Please refer to Slide 18 of Lecture 4 of Week 8

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14. Which of the following represents the applications of IoT in Livestock Barns?

a. Soil moisture sensors optimize irrigation and fertilization
b. Monitoring health parameters of animals
c. Drone-assisted mining monitoring
d. None of the above

Answer: (b)

Justification: Please refer to Slide 7 of Lecture 5 of Week 8

15.In the context of Agricultural IoT, what do we mean by LAI?

a. Leaf Area Index
b. Layout Area Increment
c. Low Achor Installation
d. None of the above

Answer: (a)

Justification: Please refer to Page 276 of the book Introduction to IoT by S Misra, A Mukherjee, A Roy by Cambridge University Press

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